

POWER BI WEEK-7

REPORT

Introduction

Week 7 focuses on Student Performance Data Cleaning and Transformation using Power BI. The Student Performance dataset contains information about students' gender, race/ethnicity, parental education, lunch type, test preparation, and their Math, Reading, and Writing scores. In this week, the dataset is explored, cleaned, and transformed using Power Query. New calculated columns such as Total Marks, Average Total, Performance Grade, and Result are created to evaluate students' academic performance. The subject scores are also unpivoted into a structured format, and DAX measures are created to calculate subject-wise averages, overall average score, and pass percentage.

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YEAR: II Year

ROLL NO: E25DA018

Figure 1: Importing the Student Performance dataset into Power BI.

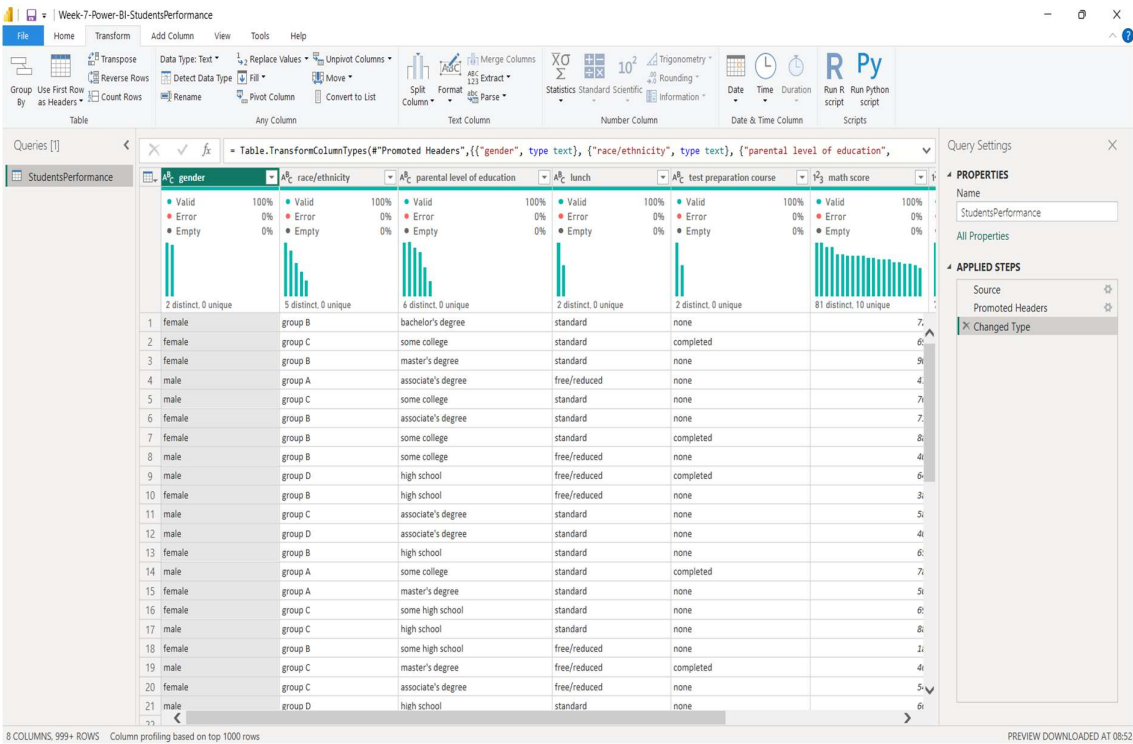


Figure 2: Checking the data types of the dataset columns.

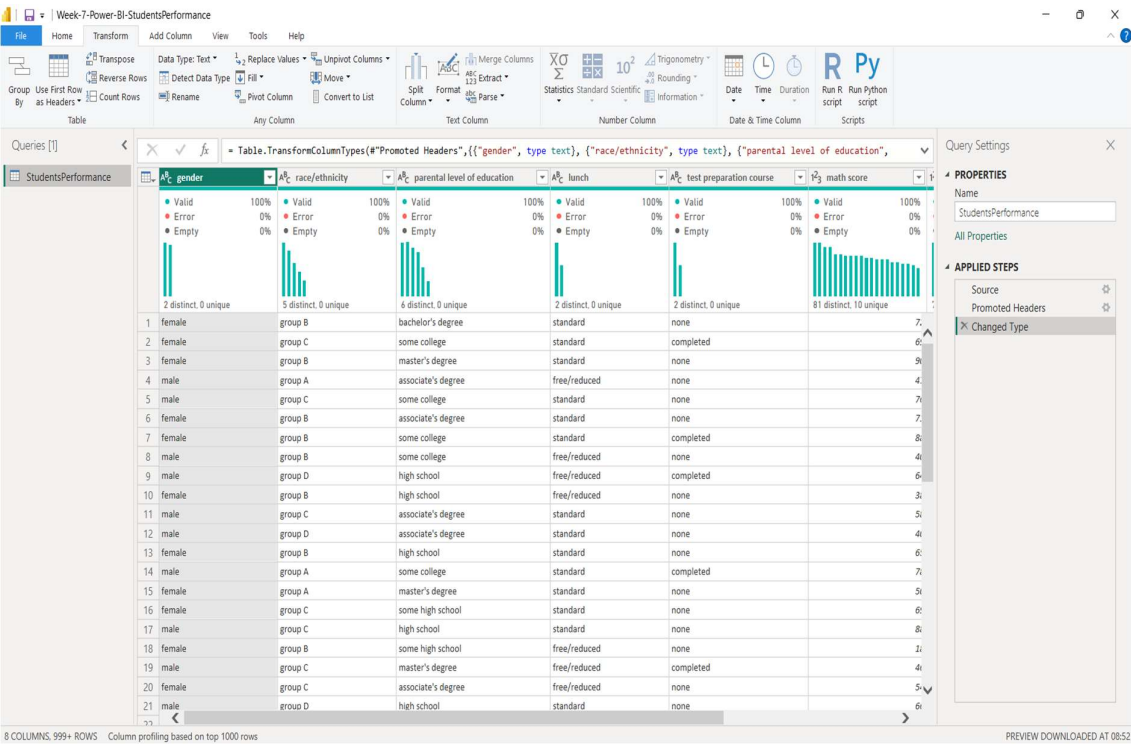


Figure 3: Standardizing the Parental Education column using Capitalize Each Word.

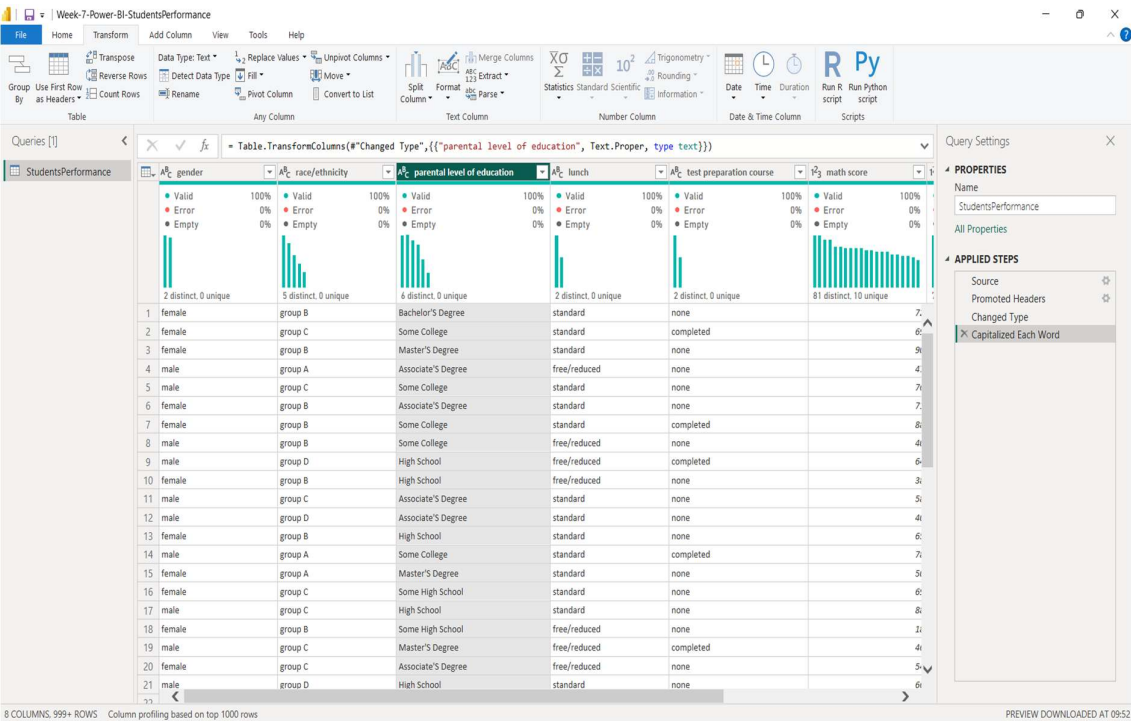


Figure 4: Creating the Total Marks column by adding Math, Reading, and Writing scores.

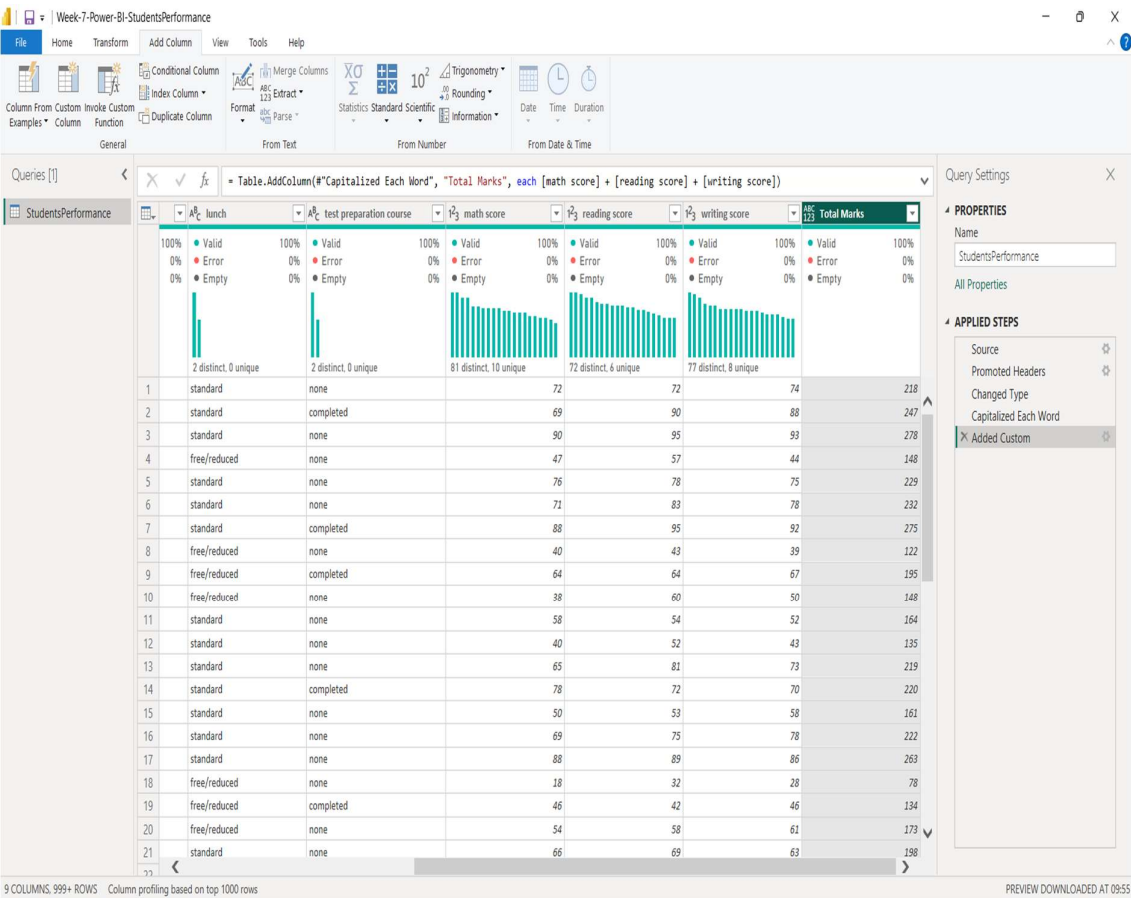


Figure 5: Creating Average Total by dividing Total Marks by three and setting the data type to Decimal Number.

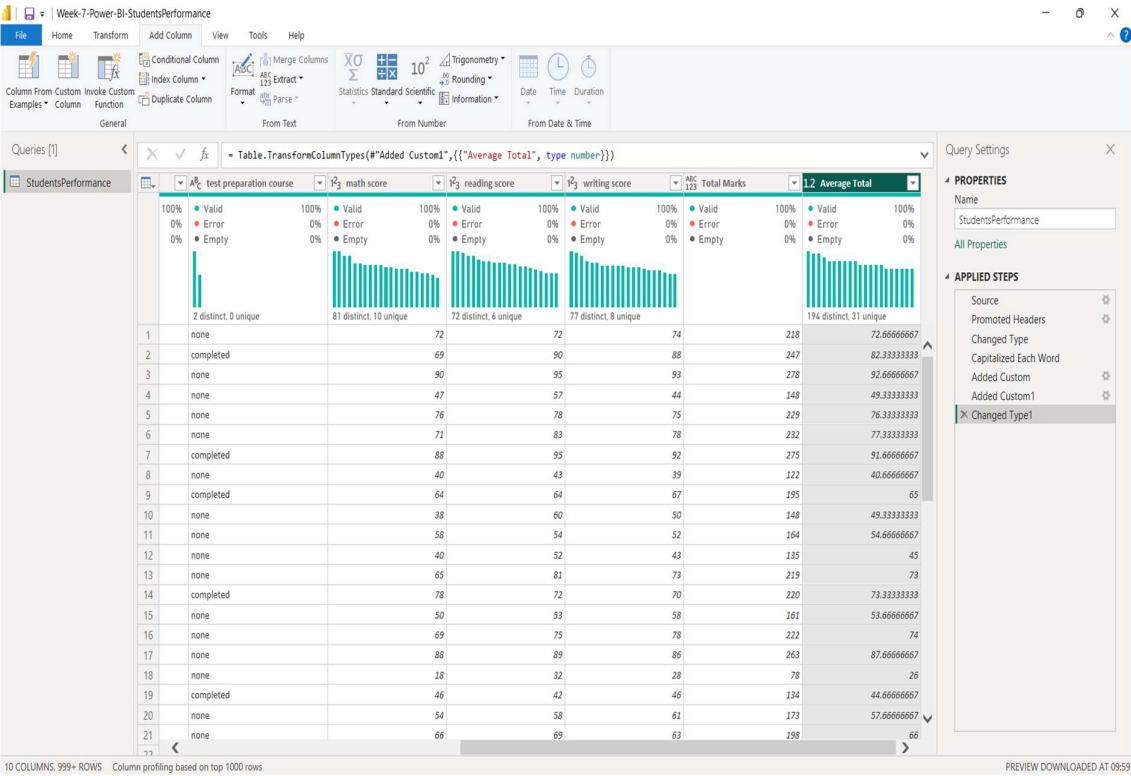


Figure 6: Creating the Performance Grade column using conditional logic based on Average Total.

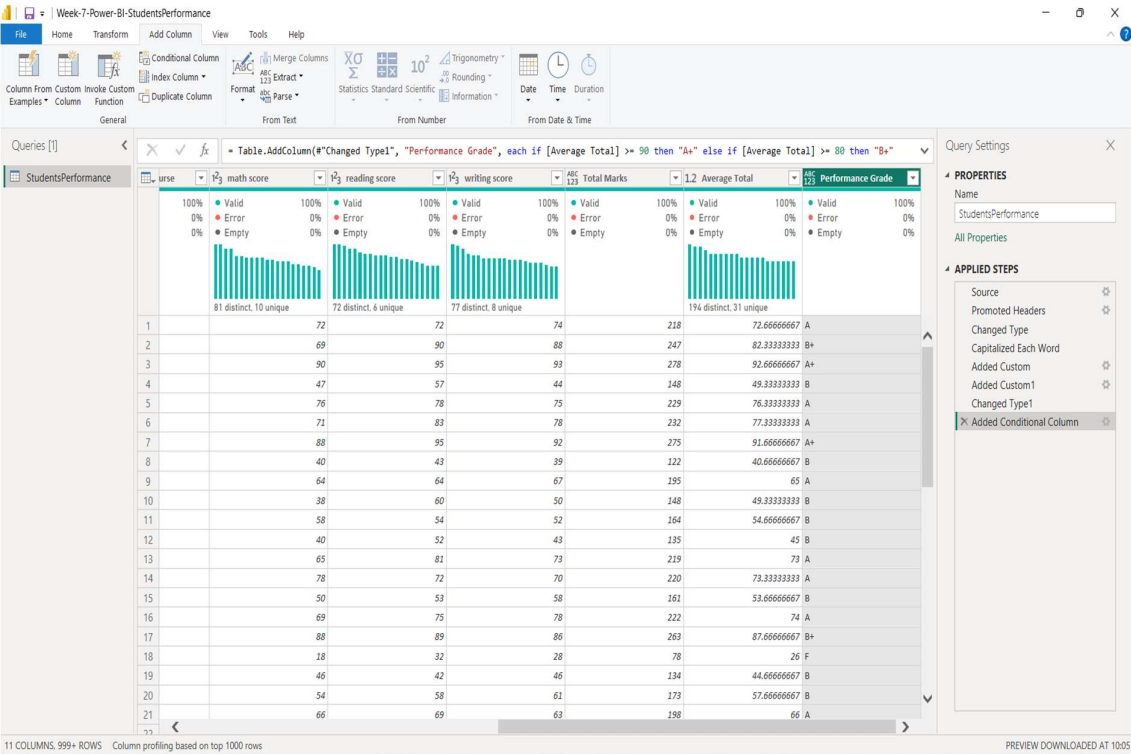


Figure 7: Creating the Result column to classify students as Pass or Fail.

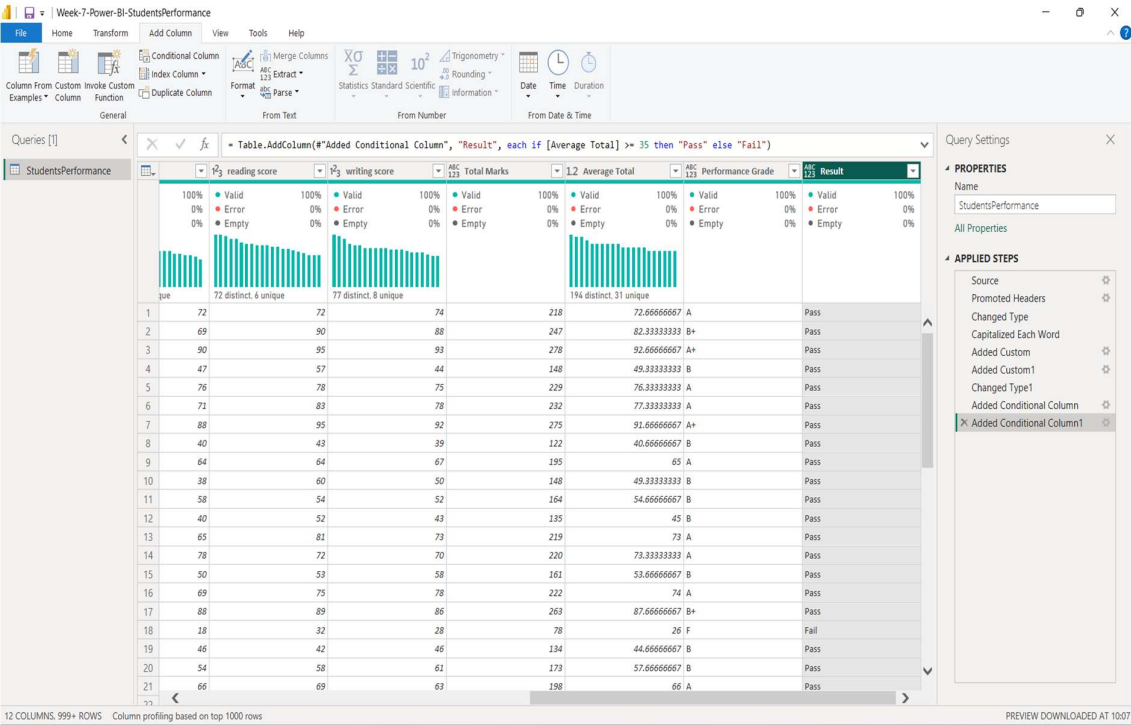


Figure 8: Creating a unique Student ID using an index column for student identification.

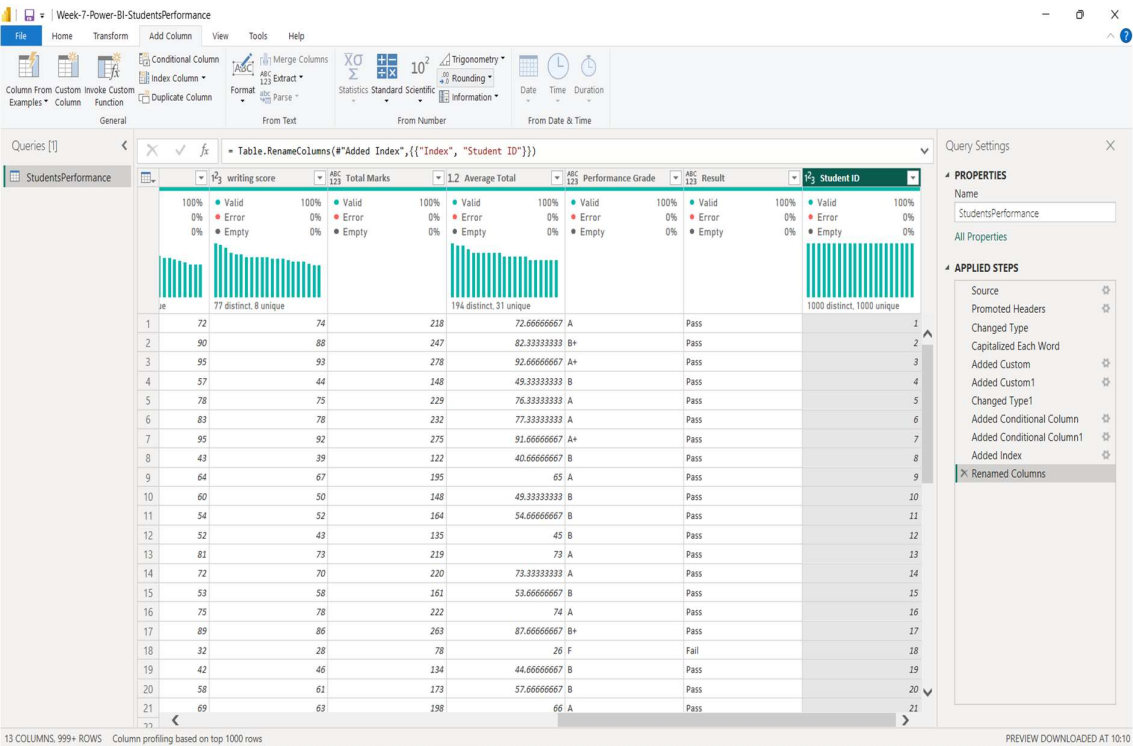


Figure 9: Duplicating the student dataset to create the Student Marks table.

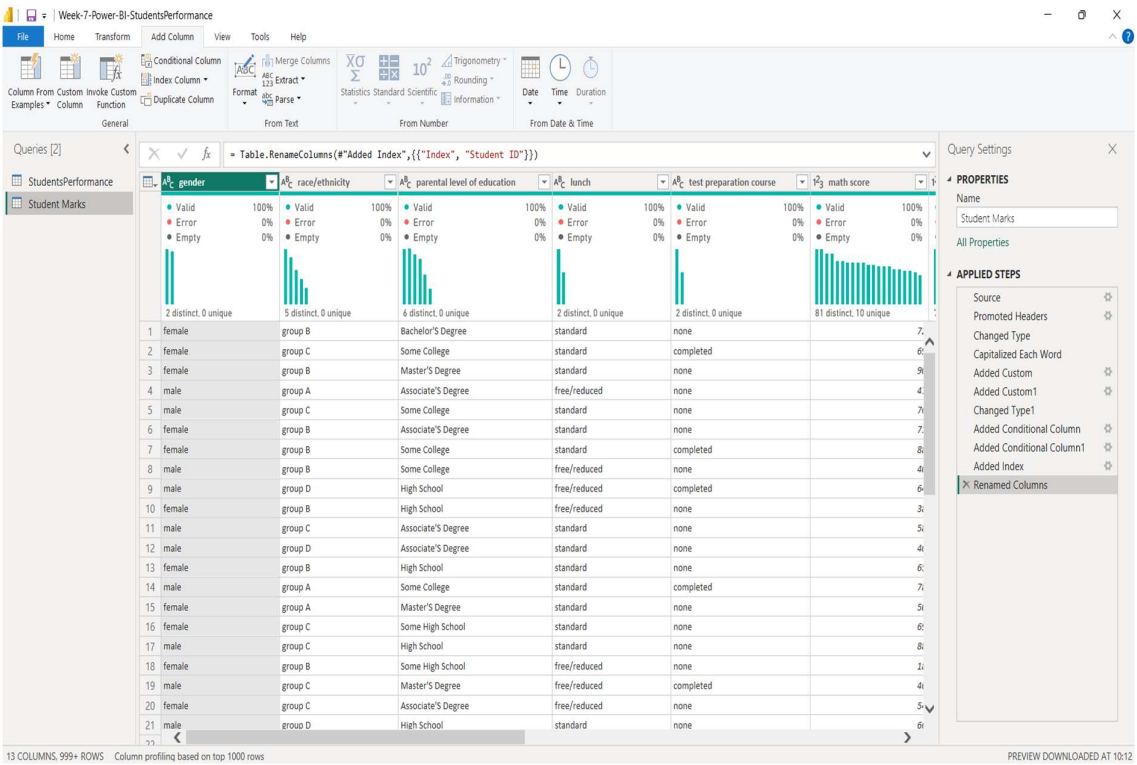


Figure 10: Preparing the Student Marks table with Student ID and the three subject score columns.

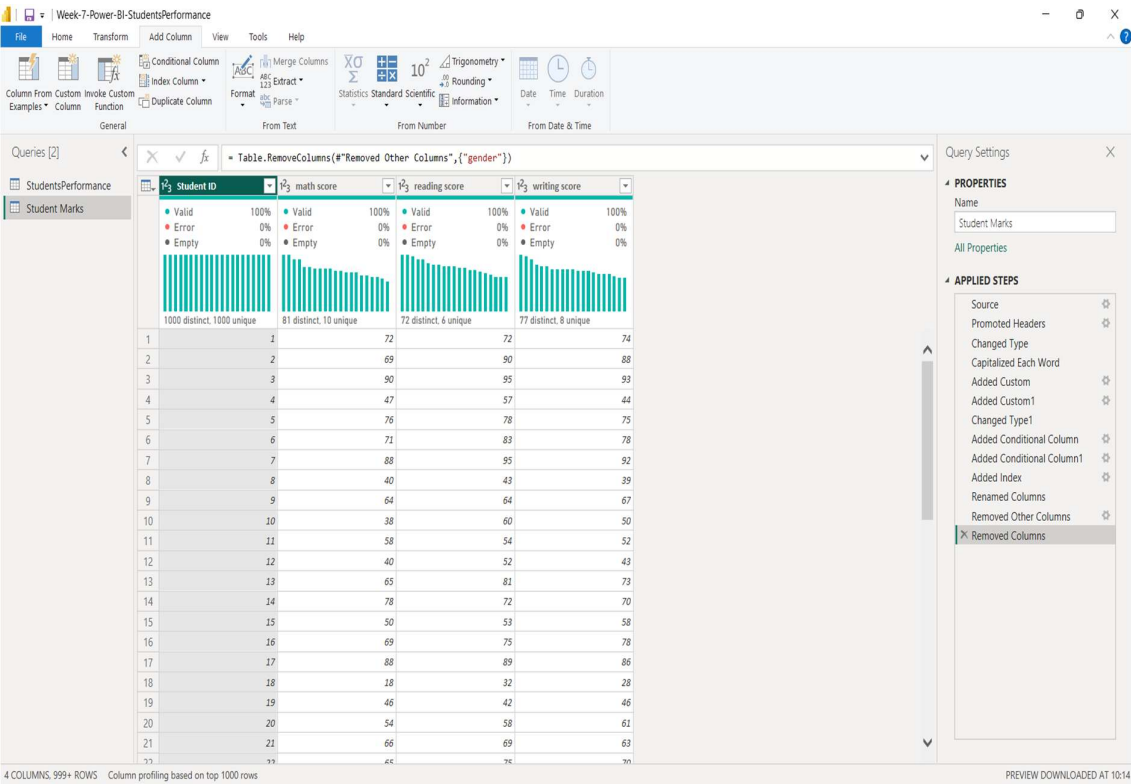


Figure 11: Unpivoting the Math, Reading, and Writing score columns into Subject and Score attributes.

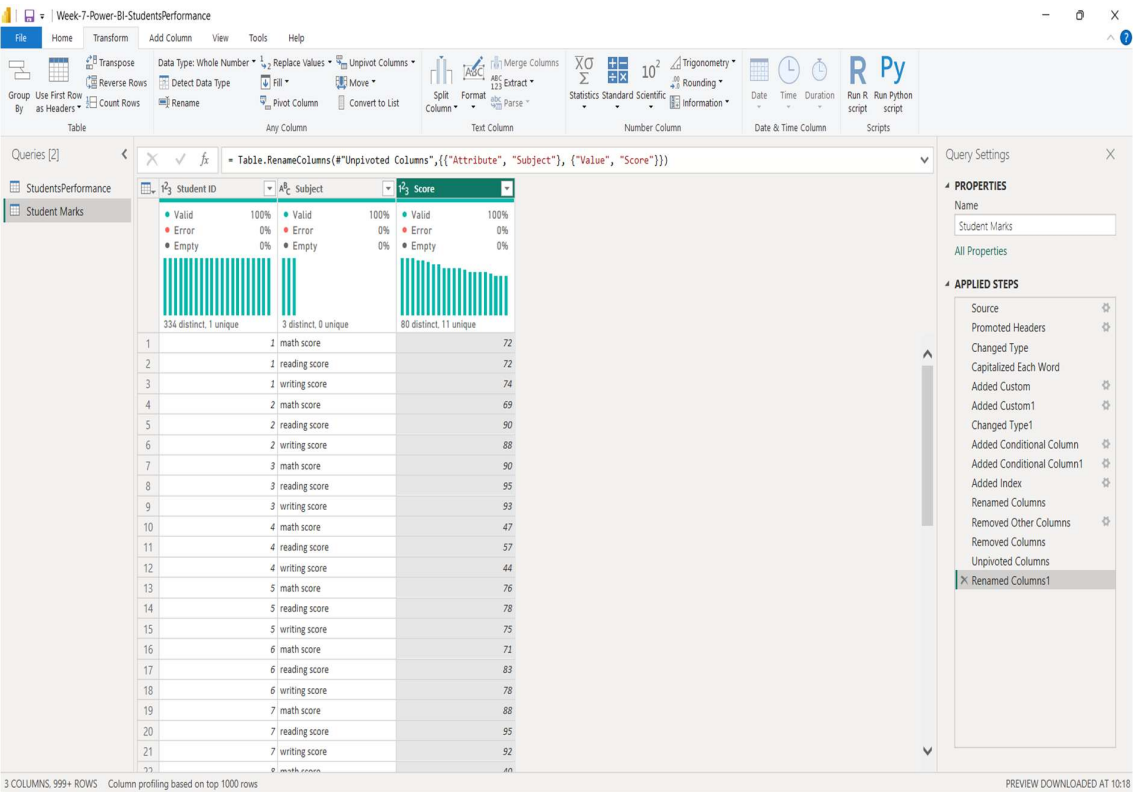


Figure 12: Applying the transformed Student Performance and Student Marks tables using Close & Apply.

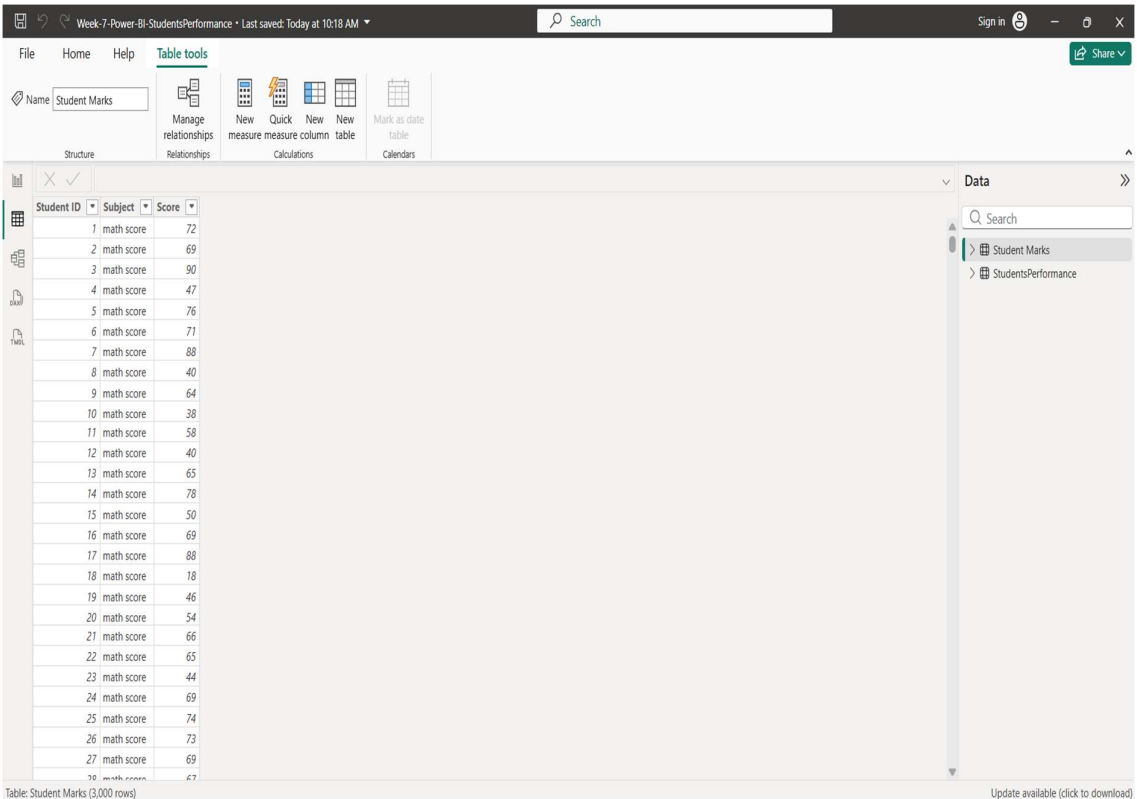


Figure 13: Creating the Average Math Score DAX measure using the StudentsPerformance table.

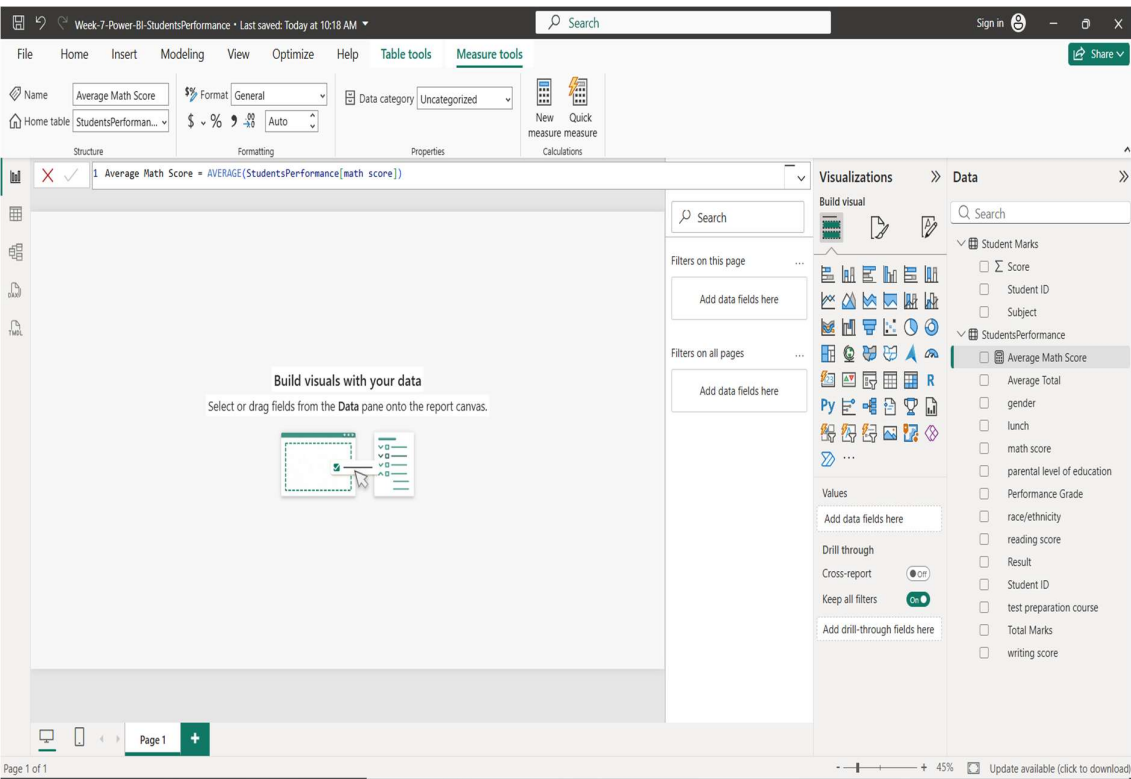


Figure 14: Creating the Average Reading Score DAX measure.

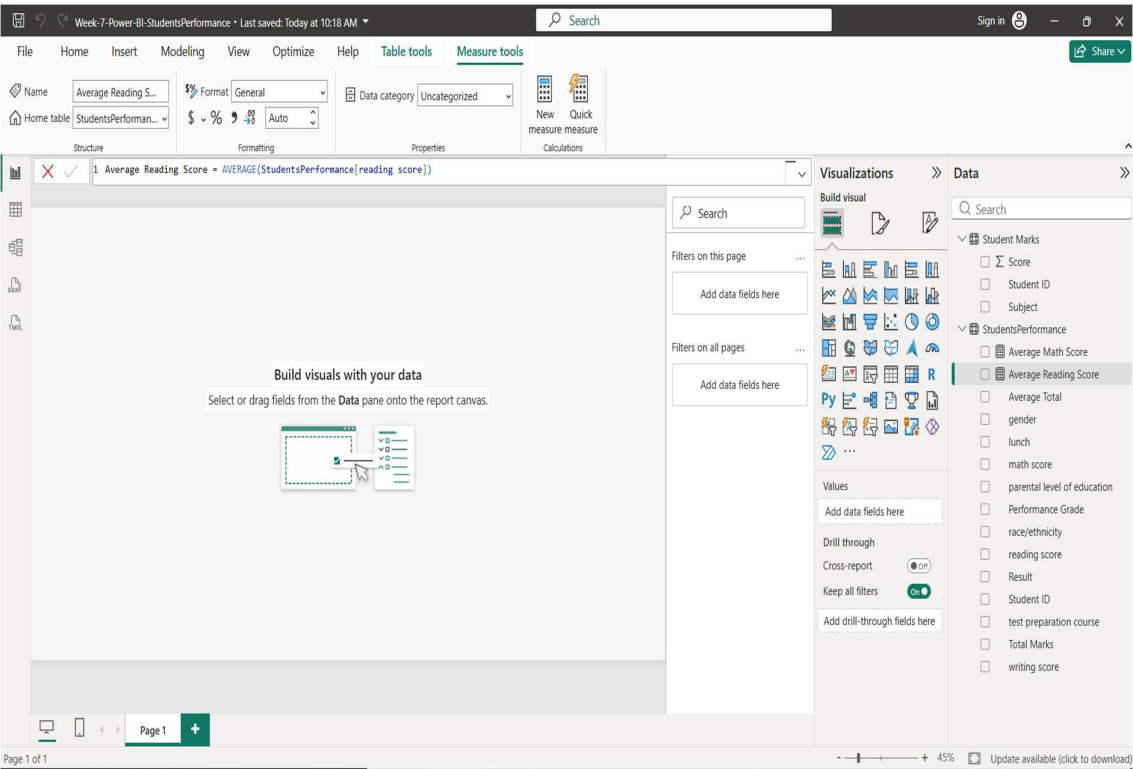


Figure 15: Creating the Average Writing Score DAX measure.

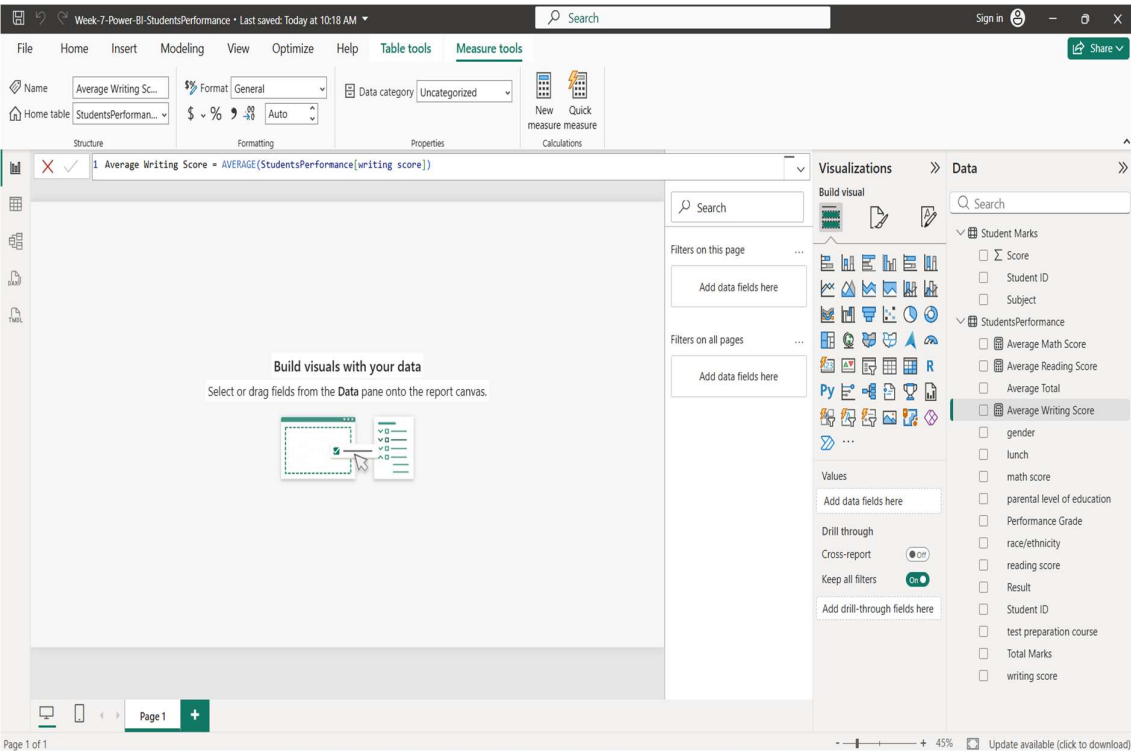


Figure 16: Creating the Overall Average Score DAX measure.

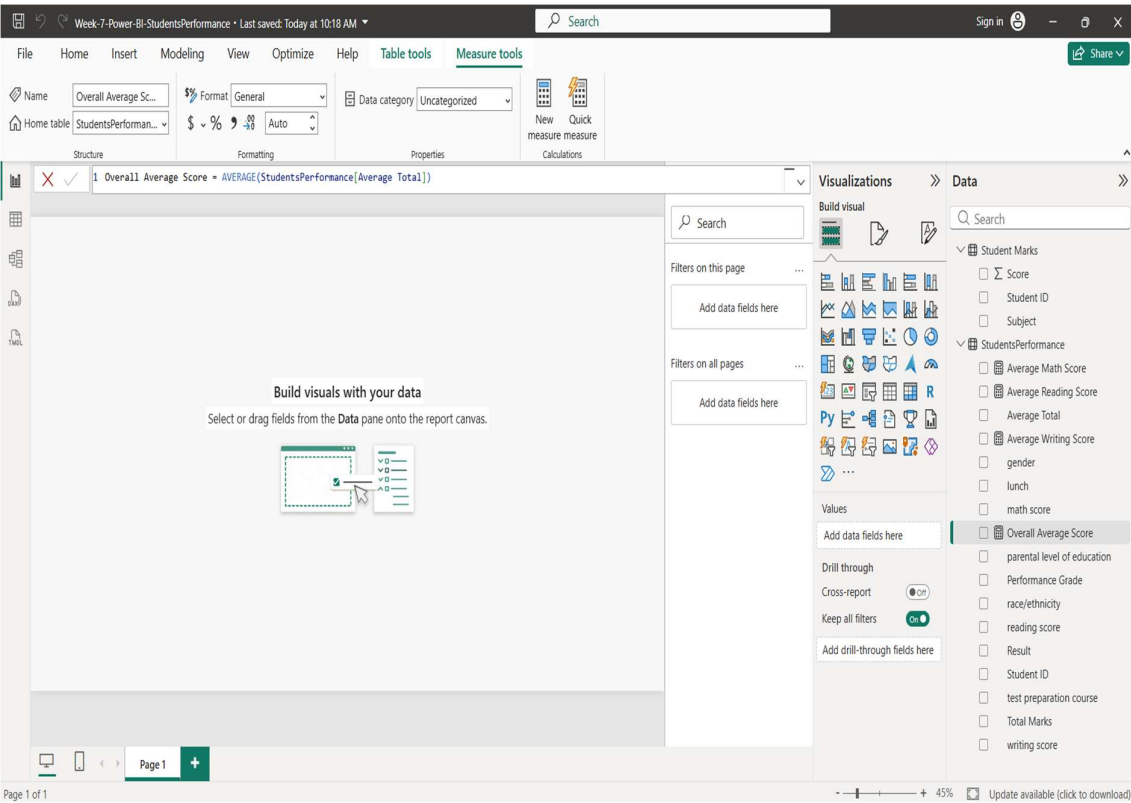


Figure 17: Creating the Overall Pass Percentage DAX measure.

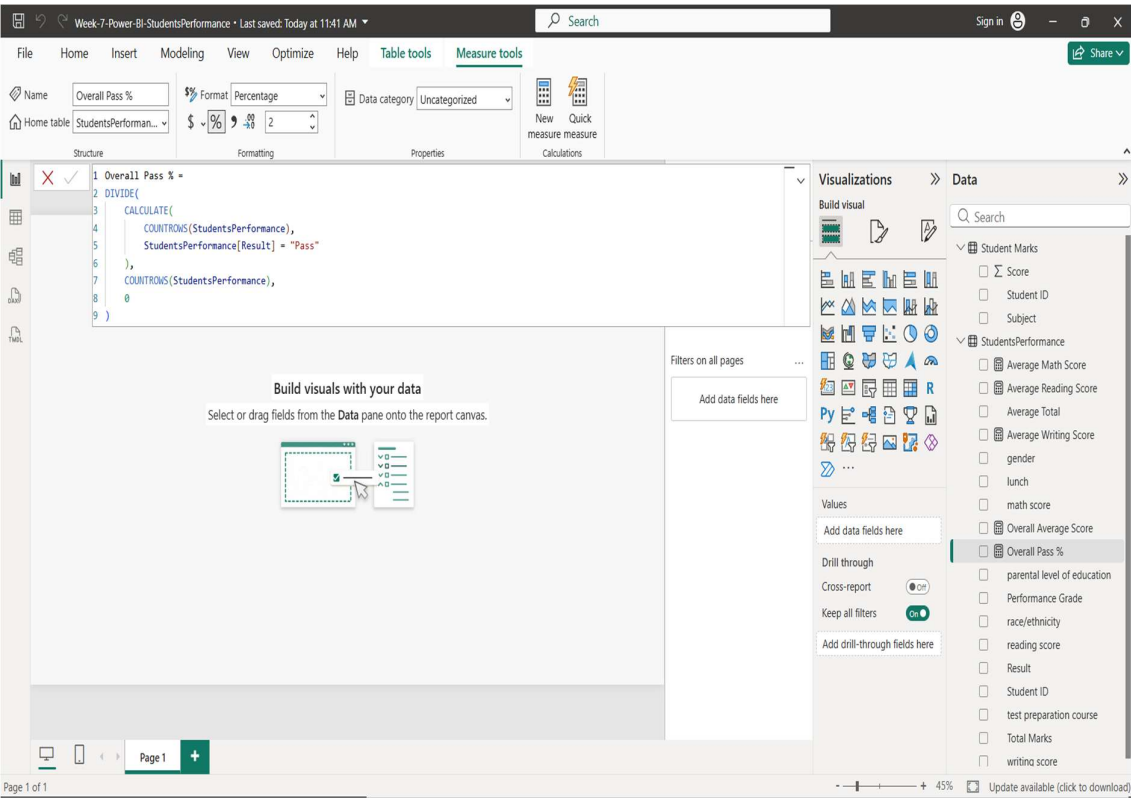


Figure 18: Verifying the relationship between StudentsPerformance and Student Marks using Student ID.

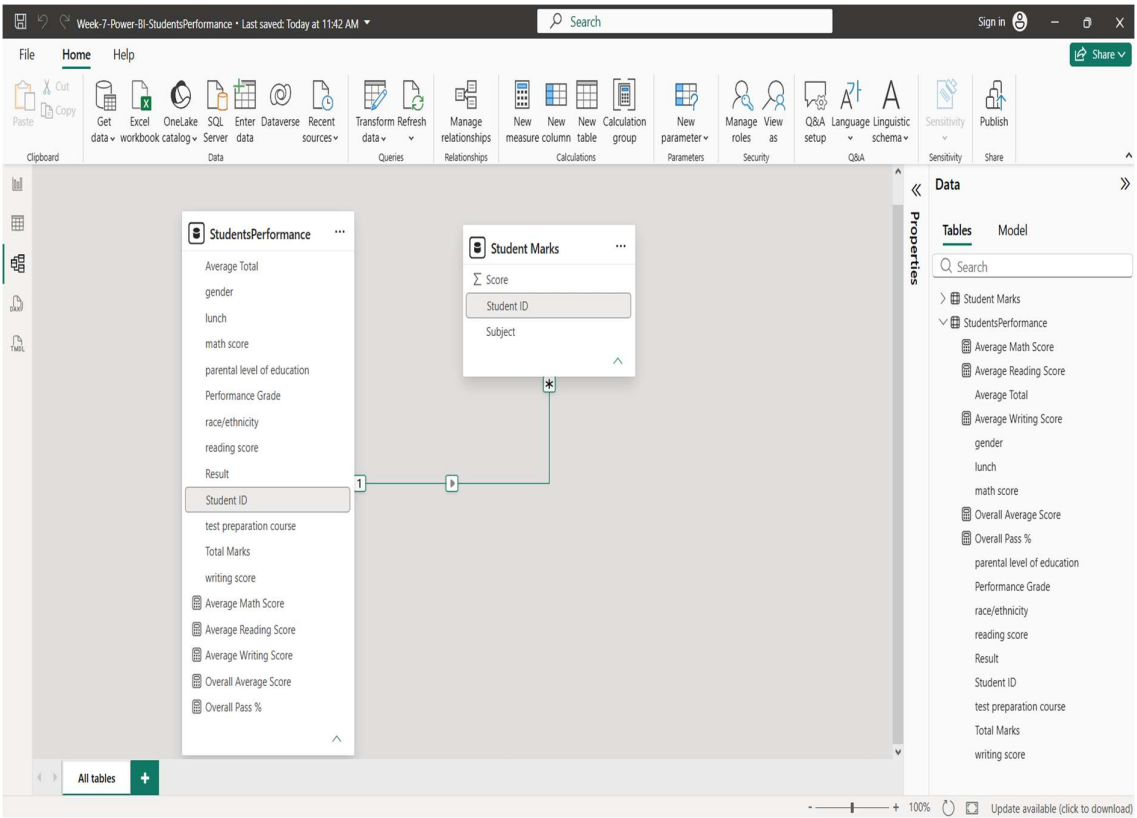


Figure 19: Validating the calculated marks, average scores, performance grades, and Pass/Fail results in the StudentsPerformance table.

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parental level of educationlunchtest preparation coursemath scorereading scorewriting scoreTotal MarksAverage TotalPerformance GradeResultStudent ID

Some Collegestandardnone76787522976.3333333333333333APass5

Some High Schoolstandardnone69757822274APass16

Some High Schoolstandardnone69737321571.666666666666667APass24

Bachelor'S Degreestandardnone67697521170.3333333333333333APass28

High Schoolstandardnone70706520568.3333333333333333APass29

Associate'S Degreestandardnone58736819966.3333333333333333APass42

High Schoolstandardnone66717621371APass48

Associate'S Degreestandardnone69807122073.3333333333333333APass64

Some Collegestandardnone60727420668.666666666666667APass68

Some Collegestandardnone73808223578.3333333333333333APass86

Bachelor'S Degreestandardnone65727421170.3333333333333333APass91

High Schoolstandardnone71797122173.666666666666667APass93

Some Collegestandardnone58677219765.666666666666667APass106

High Schoolstandardnone84777423578.3333333333333333APass116

Some Collegestandardnone55696518963APass120

Associate'S Degreestandardnone59666719264APass139

Some Collegefree/reducednone59626418561.666666666666667APass142

Associate'S Degreefree/reducednone73686620769APass148

Some Collegefree/reducedcompleted67757021270.666666666666667APass170

Associate'S Degreestandardnone63677020066.666666666666667APass174

Associate'S Degreestandardcompleted71777722575APass178

Master'S Degreefree/reducedcompleted62687520568.3333333333333333APass181

Associate'S Degreefree/reducedcompleted65676519765.666666666666667APass186

Master'S Degreestandardcompleted69848523879.3333333333333333APass195

Some High Schoolfree/reducedcompleted59696519364.3333333333333333APass197

Associate'S Degreestandardcompleted67848623779APass201

Associate'S Degreestandardnone69776921571.666666666666667APass203

Table: StudentsPerformance (1,000 rows)

Update available (click to download)

Conclusion

The Week 7 activity successfully transformed the Student Performance dataset using Power BI and Power Query. The dataset was inspected, data types were verified, and the Parental Education values were standardized. New calculated columns including Total Marks, Average Total, Performance Grade, and Result were created to evaluate students' academic performance. A unique Student ID was added, and a duplicate Student Marks table was created for subject-level analysis. The Math, Reading, and Writing scores were unpivoted into Subject and Score columns. Finally, DAX measures were created for subject-wise averages, overall average score, and overall pass percentage, and the relationship between the two tables was validated.